# Experiment 40: Soccer Player Data Analysis

**Aim:** To analyze soccer player data using Pandas and visualize player positions using a bar chart.

**Algorithm:**

1. Import Pandas and Matplotlib.
2. Create a soccer player dataset.
3. Save the dataset as a CSV file.
4. Read the CSV file into a DataFrame.
5. Find the top five players by goals and salary.
6. Calculate average age.
7. Display players above average age.
8. Draw a bar chart for player positions.

**Code:**

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Name": ["A", "B", "C", "D", "E", "F"],
    "Age": [24, 28, 30, 22, 27, 32],
    "Position": ["Forward", "Midfielder", "Defender",
                 "Forward", "Defender", "Midfielder"],
    "Goals": [20, 10, 5, 25, 8, 12],
    "Salary": [50000, 40000, 30000, 60000, 35000, 55000]
}

df = pd.DataFrame(data)

df.to_csv("players.csv", index=False)
df = pd.read_csv("players.csv")

print("Top 5 Players by Goals:")
```

```
print(df.nlargest(5, "Goals")[["Name", "Goals"]])

print("Top 5 Players by Salary:")
print(df.nlargest(5, "Salary")[["Name", "Salary"]])

avg = df["Age"].mean()
print("Average Age:", avg)

print("Players Above Average Age:")
print(df[df["Age"] > avg]["Name"])

df["Position"].value_counts().plot(kind="bar")
plt.xlabel("Position")
plt.ylabel("Number of Players")
plt.title("Players by Position")
plt.show()
```

**Input:**

Player data:

A, 24, Forward, 20, 50000

B, 28, Midfielder, 10, 40000

C, 30, Defender, 5, 30000

D, 22, Forward, 25, 60000

E, 27, Defender, 8, 35000

F, 32, Midfielder, 12, 55000

**Output:**

Top 5 Players by Goals:

D 25

A 20

F 12

B 10

E 8

Top 5 Players by Salary:

D 60000

F 55000

A 50000

B 40000

E 35000

Average Age: 27.17

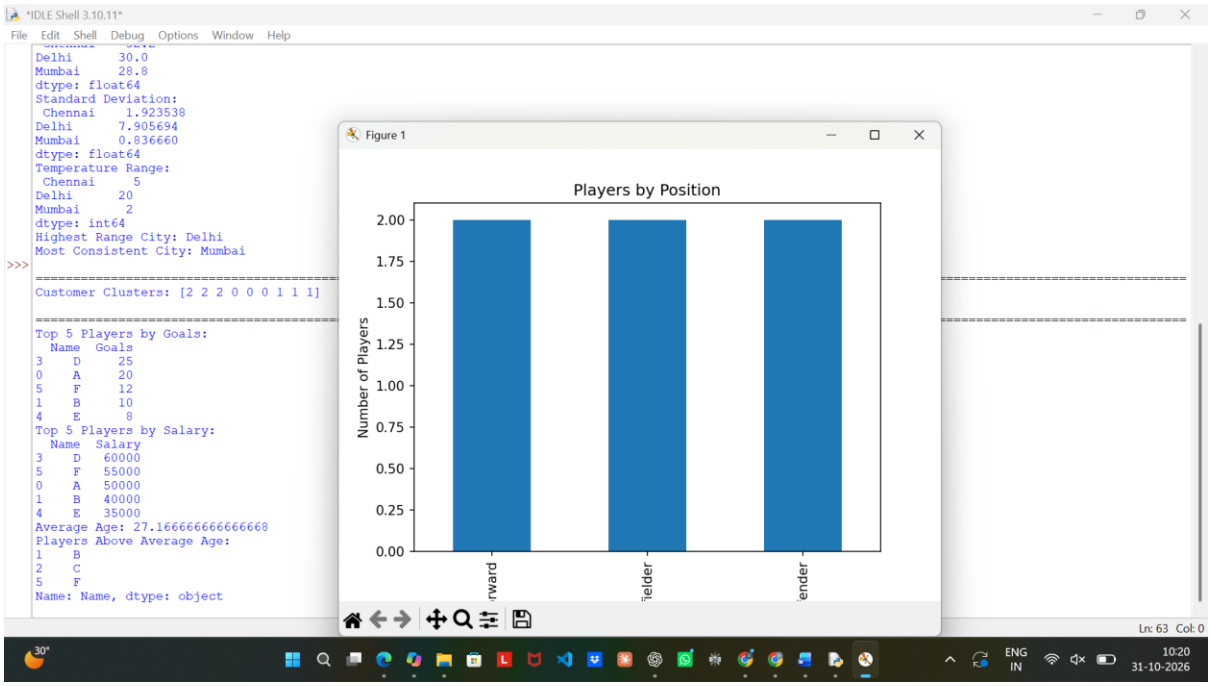
Players Above Average Age:

B

C

F

A bar chart showing the number of players in each position will appear.




**Result:** Soccer player data was analyzed, top players were identified, and player positions were visualized.